

For the full score it is enough to solve 3 problems.

1. Check the convergence of the series

$$\sum_{k=1}^{\infty} \left(\sqrt{k+1} - \sqrt{k} \right)^{-1/2} \ln \frac{k-1}{k+1}.$$

Solution. Notice that

$$\left(\sqrt{k+1} - \sqrt{k} \right)^{-1/2} = \frac{1}{\sqrt{\sqrt{k+1} - \sqrt{k}}} = \sqrt{\frac{\sqrt{k+1} + \sqrt{k}}{k+1-k}} \sim \sqrt{2}k^{1/4}$$

and

$$\ln \frac{k-1}{k+1} = \ln \left(1 - \frac{2}{k+1} \right) \sim -\frac{2}{k+1} \sim -\frac{2}{k}.$$

Consequently,

$$a_k = \left(\sqrt{k+1} - \sqrt{k} \right)^{-1/2} \ln \frac{k-1}{k+1} \sim -\frac{\sqrt{2}}{k^{3/4}}$$

and the series $\sum_{k=1}^{\infty} a_k$ diverges since $3/4 < 1$.

2. Check the uniform convergence of the functional sequence

$$f_n(x) = e^{n(x-1)}$$

on $E = (0, 1)$.

Solution. First, let's find the pointwise limit

$$f_n(x) = e^{n(x-1)} \rightarrow 0 = f(x), \quad n \rightarrow \infty$$

for every $x \in (0, 1)$.

To check the uniform convergence consider

$$\sup_{x \in (0,1)} |f_n(x) - f(x)| = 1.$$

Consequently, the series doesn't converge absolutely.

3. Check the uniform convergence of the series

$$\sum_{k=1}^{\infty} \frac{x}{1 + n^4 x^3}$$

on $E = (0, +\infty)$.

Solution. Let $u_n(x) = \frac{x}{1+n^4x^3}$. To find $\sup u_n$ consider

$$u'_n(x) = \frac{1 + n^4 x^3 - x(3n^4 x^2)}{(1 + n^4 x^3)^2} = \frac{1 - 2n^4 x^3}{(1 + n^4 x^3)^2}.$$

Then $u'_n(x) = 0$ iff $x = \frac{1}{\sqrt[3]{2}n^{4/3}}$ and since $u_n(0) = \lim_{x \rightarrow \infty} u_n(x) = 0$

$$\sup_{x \in (0, \infty)} u_n(x) = u_n\left(\frac{1}{\sqrt[3]{2}n^{4/3}}\right) = \frac{2}{3n^{4/3}}.$$

Consequently, the series is convergent by Weierstrass M-test .

4. Find Maclaurin series of the function

$$f(x) = \frac{1}{4} \ln \frac{1+x}{1-x} + \frac{1}{2} \arctan x.$$

Solution. Consider

$$\begin{aligned} f'(x) &= \frac{1}{4} \left(\frac{1}{1+x} + \frac{1}{1-x} \right) + \frac{1}{2} \frac{1}{1+x^2} = \\ &= \frac{1}{2} \left(\frac{1}{1-x^2} + \frac{1}{1+x^2} \right) = \frac{1}{1-x^4} = \sum_{k=0}^{\infty} x^{4k}. \end{aligned}$$

Consequently,

$$f(x) = f(0) + \sum_{k=0}^{\infty} \frac{x^{4k+1}}{4k+1} = \sum_{k=0}^{\infty} \frac{x^{4k+1}}{4k+1}.$$

5. Find the radius of convergence and the sum of the power series

$$\sum_{k=1}^{\infty} k(k+2)x^k.$$

Solution. First

$$R^{-1} = \lim \sqrt[k]{k(k+2)} = 1.$$

Also

$$\begin{aligned} f(x) &= \sum_{k=1}^{\infty} k(k+2)x^k = \sum_{k=1}^{\infty} k(k+2)x^k = \\ &= \sum_{k=1}^{\infty} (k+1)(k+2)x^k - \sum_{k=1}^{\infty} (k+1)x^k - \sum_{k=1}^{\infty} x^k = \\ &= \sum_{k=1}^{\infty} (x^{k+2})'' - \sum_{k=1}^{\infty} (x^{k+1})' - \sum_{k=1}^{\infty} x^k = \\ &= \left(\sum_{k=1}^{\infty} x^{k+2} \right)'' - \left(\sum_{k=1}^{\infty} x^{k+1} \right)' - \frac{x}{1-x} = \\ &= \left(\frac{x^3}{1-x} \right)'' - \left(\frac{x^2}{1-x} \right)' - \frac{x}{1-x} = \frac{x(3-x)}{(x-1)^3}, \quad x \in (-1, 1). \end{aligned}$$

Solution 2.

$$\begin{aligned} f(x) &= \sum_{k=1}^{\infty} k(k+2)x^k = \sum_{k=1}^{\infty} k(k+2)x^k = x \sum_{k=1}^{\infty} k(k+2)x^{k-1} = \\ & x \left(\sum_{k=1}^{\infty} (k+2)x^k \right)' = x \left(x \sum_{k=1}^{\infty} (k+2)x^{k-1} \right)' = \\ & x \left(x \left(\sum_{k=1}^{\infty} x^k \right)' + 2 \sum_{k=1}^{\infty} x^k \right)' \end{aligned}$$